

Smart Medicine Reminder System for DVAGO

Project Proposal



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1. Introduction

Managing daily medications is a challenge for many patients, especially those with chronic illnesses. While DVAGO offers reliable online pharmacy services, it currently lacks a built-in reminder system. This project, developed with DVAGO's approval, presents a smart medicine reminder prototype with alerts, voice notifications, and refill suggestions. It also features an AI-based tool to recognize medicine names through image scanning for easier setup.

2. Objective

The primary objective of this project is to develop a **smart medicine reminder system for DVAGO** that assists users in managing medication intake through scheduled notifications and optional voice alerts. The system also aims to simplify the reminder setup process by integrating an **AI-based image scanning tool** that identifies medicine names from boxes or wrappers. This solution focuses on improving user convenience, adherence, and engagement with digital healthcare services.

3. Problem Description

Many patients miss doses or refills due to the absence of timely reminders, which can disrupt treatment and reduce engagement with pharmacy platforms. DVAGO currently lacks a built-in tool for structured medication tracking. This project proposes a modular, user-friendly reminder system that notifies users via push and voice alerts. It also includes an AI-based image scanning feature to identify medicine names from packaging, making reminder setup quicker and more accessible.

4. Comprehensive Literature Review

Digital medicine reminder apps, such as Medisafe³ and MyTherapy⁴, have been shown to improve patient adherence by over 20%¹ in countries including the USA, UK, and Germany. These platforms offer scheduled alerts, refill reminders, and simplified tracking to help users stay consistent with their prescriptions.

However, such features are still uncommon in Pakistan's e-pharmacy platforms. This presents DVAGO with an opportunity to lead the local market by offering a **locally adapted, smart reminder system**.

Our proposed solution is **lightweight, mobile-friendly**, and easy to use. It supports users with **push notifications, voice alerts**, and a new **AI-based scanning feature** for quick reminder setup, without disrupting the digital experience².

5. Methodology

This project follows the **Agile development methodology**, allowing iterative development and continuous testing through well-defined sprints. Each sprint focuses on implementing and refining specific system features to ensure flexibility and quality improvement throughout the development cycle. Key components of the methodology include:

- **Reminder Logic:** Automatically schedules alerts based on user-defined dosage timings and medicine duration.
- **Notification System:** Implements Firebase Cloud Messaging (FCM) to deliver real-time push notifications.
- **Voice Alert Integration:** Utilizes platform-based Text-to-Speech (TTS) APIs to provide audio reminders.
- **User Interface:** Developed using **React Native** to ensure a consistent and responsive mobile experience on Android and iOS platforms.
- **AI-Based Medicine Scanner:** An optional feature that uses OCR and image recognition to identify medicine names from packaging, simplifying reminder setup.

The development process is divided into iterative sprints, with each cycle involving design, implementation, and testing to ensure system reliability and usability.

6. Project Scope

Included:

- Designing and developing a reminder system for DVAGO's digital platforms
- Notification and voice-based alerts for timely medicine intake
- History tracking and user-configurable reminder settings
- AI-based medicine recognition via image scanning for easier reminder setup

Out of Scope:

- Live integration, backend access, or commercial deployment without DVAGO's approval

7. Feasibility Study

Risks Involved

- Users may ignore or disable reminder notifications
- Voice alert functionality may have limited support on older devices
- AI-based medicine recognition accuracy may vary with low-quality images

Resources Required

- Firebase services for authentication and notifications
- MongoDB for data storage
- Android and iOS devices for testing
- Development laptops with required tools
- OCR or image recognition libraries for AI scanning

8. Solution Application Areas

- Individuals taking daily or long-term medications
- Elderly patients and users managing chronic illnesses
- Users who face difficulty in manually entering medicine names
- DVAGO's initiative to enhance digital healthcare services through smart, value-added solutions

9. Tools/Technology

Frontend: React Native

Backend: Node.js

Database: Firebase, MongoDB

APIs: Firebase Cloud Messaging (Push Notifications), Google Vision API (AI-based Medicine Recognition), Text-to-Speech

Design Tools: Figma, Canva

Hosting & Version Control: Firebase Hosting, GitHub

10. Responsibilities of the Team Members

Task / Module	Abdul Ghani	Syed Anas	Supervisor	External Supervisor (DVAGO)
Requirement Gathering	R	R	C	C
UI/UX Design	A	R	C	I
Backend Logic (Reminder Scheduler)	R	A	C	I
Notification Integration (FCM)	R	R	C	I
Voice Alert Feature (TTS)	R	A	C	I
AI-Based Medicine Recognition	R	A	C	I
Purchase History Logic	R	A	C	I
Testing & Bug Fixing	A	R	C	C
Final Report & Documentation	R	A	C	I
Presentation Preparation	A	R	C	I
Academic Compliance (Format, Scope)	R	R	A	I

R = Responsible (the one who does the work)

A = Accountable (the one ultimately answerable)

C = Consulted (someone who provides input)

I = Informed (kept up to date on progress)

10. Planning

Task	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY
Requirement Gathering											
UI Design & AI Setup											
Backend & Notification Setup											
Voice Alerts & Testing											
Final Report & Presentation											

11. References

1. "Improving Medication Adherence with Mobile Apps" - JMIR, 2022
2. Firebase Docs: Cloud Messaging
3. Mozilla Web Speech API Documentation
4. Medisafe App Features Overview
5. WHO Report on Digital Health in Developing Countries
6. ¹ Morrison et al., *Journal of Medical Internet Research*, 2021
7. ² World Health Organization, *Adherence to Long-term Therapies*, 2019
8. ³ Medisafe <https://www.medisafe.com>
9. ⁴ MyTherapy <https://www.mytherapyapp.com>